

PROJECT SYNOPSIS PRESENTATION

Hate Speech Detection

Using Fine-Tuned MuRIL Transformer Model

“Detect it. Classify it. Prevent it.”

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NATURE OF THE PROJECT

A Machine Learning & NLP Application

Domain: Artificial Intelligence, Natural Language Processing, Machine Learning, Automated Text Moderation

Nature: Automated text classification system that learns contextual relationships instead of relying on fixed keyword lists

TECHNOLOGY STACK

MuRIL

google/muril-base-cased pre-trained Transformer

Hugging Face Transformers

Model + tokenizer implementation

PyTorch

Model training and fine-tuning

Pandas / NumPy

Data processing

Scikit-learn

Data splitting & evaluation

Streamlit

Interactive web application

Objective: demonstrate the complete ML pipeline — from dataset preparation and Transformer fine-tuning to evaluation and real-time inference.

THE PROBLEM

Statement of the Problem

Keyword-based moderation fails when meaning depends on context, language variation, or code-mixed text.

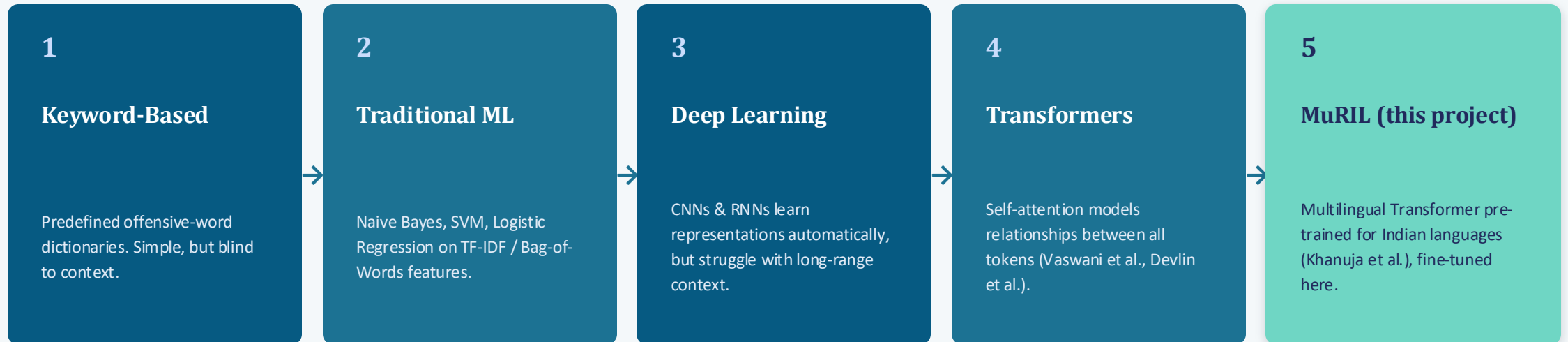
"I hate this movie because the story was terrible."

The word "hate" is present — but this sentence is not hate speech. Context matters more than keywords.

MAJOR PROBLEMS

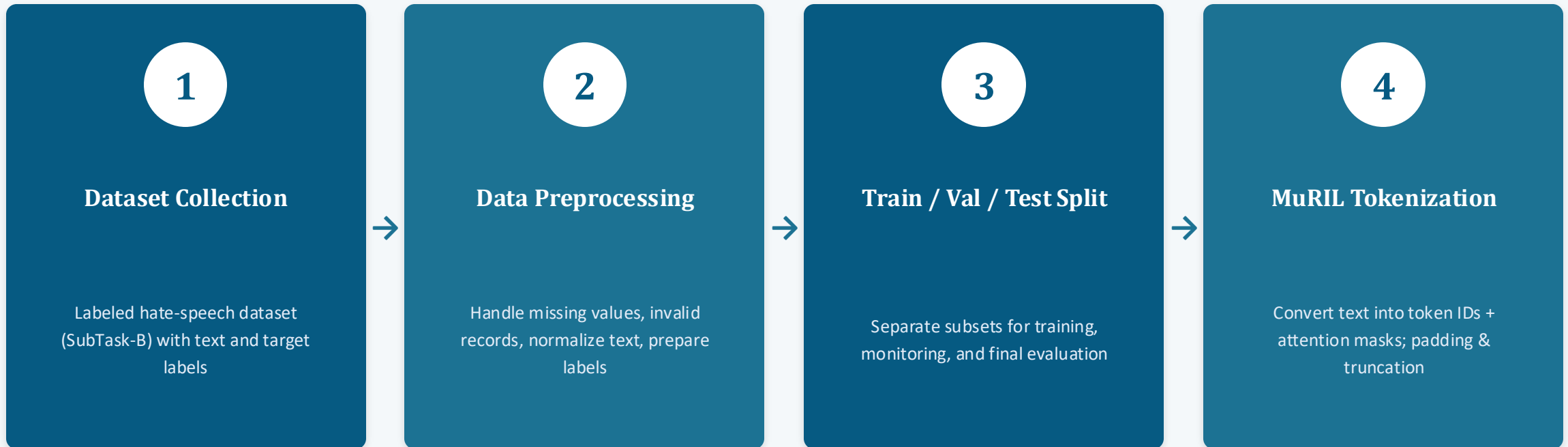
- Large volumes of online text make manual moderation difficult
- Hate-related expressions depend heavily on context
- Multilingual, Indian-language content adds further NLP challenges
- Keyword-based systems produce false positives and false negatives
- Social-media text is informal and frequently code-mixed
- A scalable, automated classification mechanism is required

Evolution of Text Classification Approaches



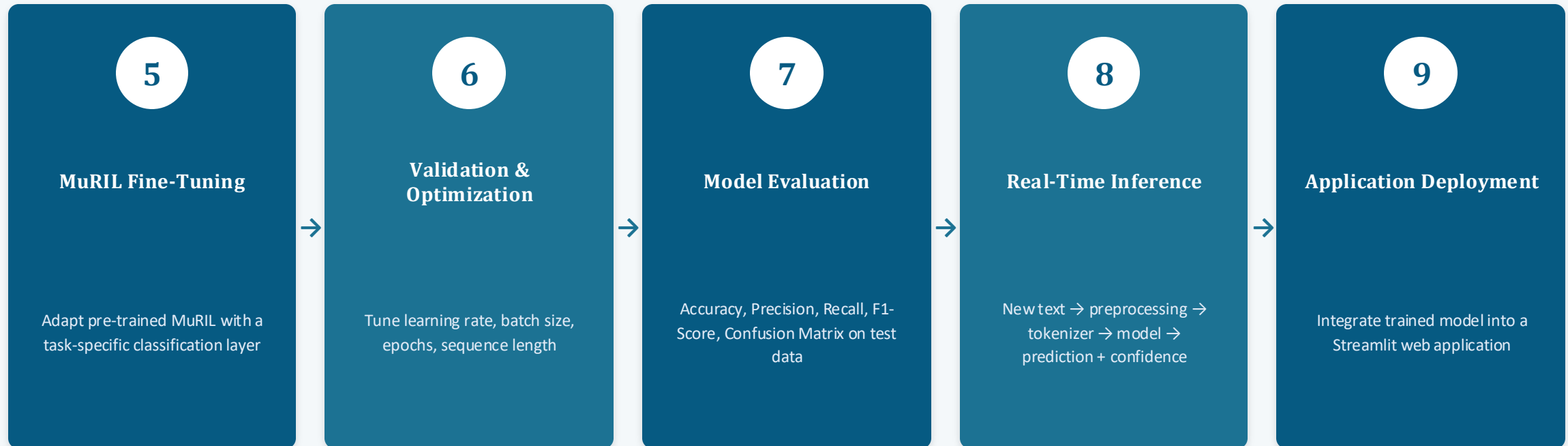
Research Gap: Most moderation systems still depend on keywords or manually engineered features. This project applies a pre-trained, multilingual, Indian-language Transformer (MuRIL) to hate speech detection and takes it beyond experimentation into a working Streamlit application.

Data Preparation Pipeline

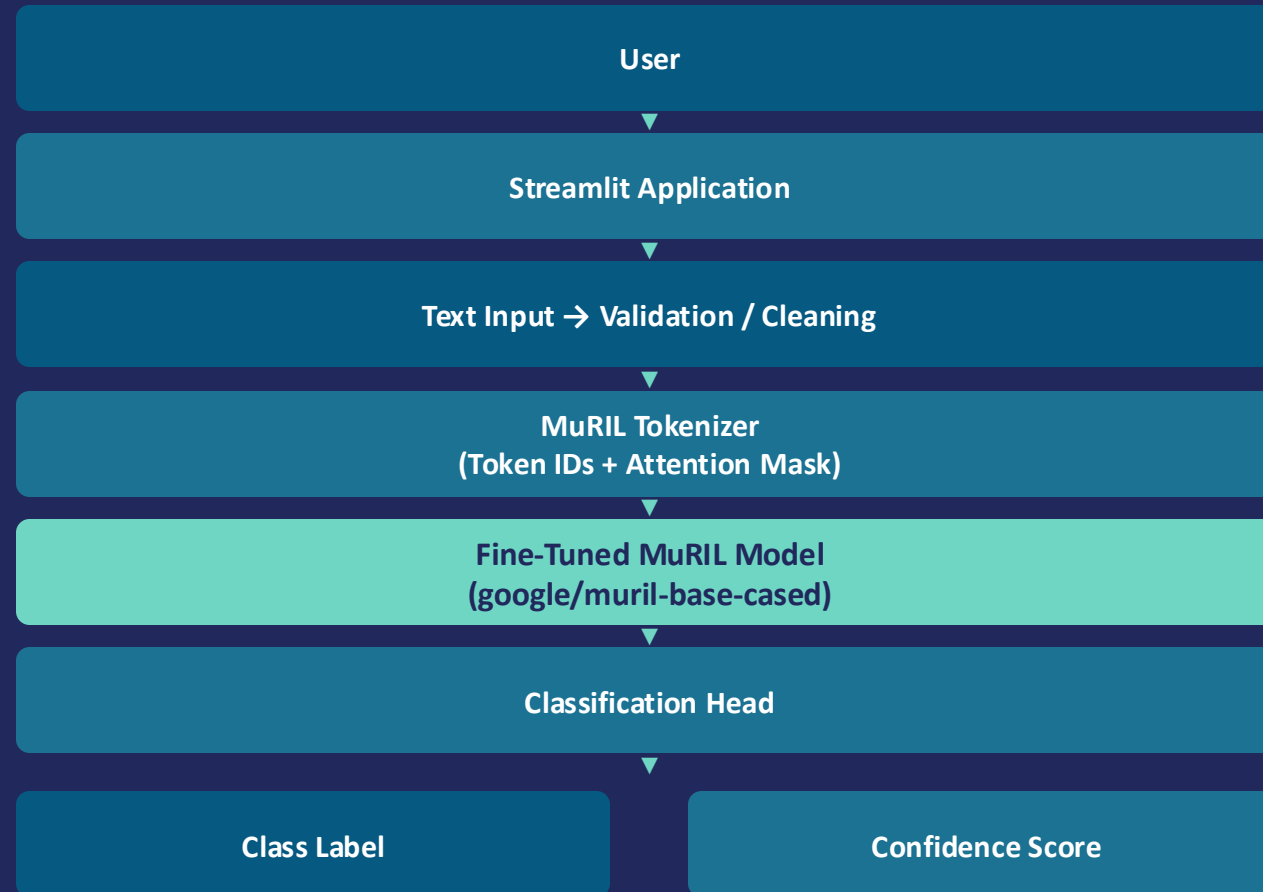


Continued in Part 2: fine-tuning, validation, evaluation, inference, and deployment.

Training, Evaluation & Deployment



Proposed Application Architecture



REQUIREMENTS

Software & Hardware Requirements

SOFTWARE

Language	Python 3.x
DL Framework	PyTorch
NLP Framework	Hugging Face Transformers
Model	google/muril-base-cased
Data Tools	Pandas, NumPy
Evaluation	Scikit-learn
Web App	Streamlit
Dev Tools	VS Code, Jupyter, Git, GitHub

HARDWARE

Processor	Intel i5/i7 or AMD Ryzen equivalent or better
RAM	8 GB minimum, 16 GB+ recommended
GPU	CUDA-compatible NVIDIA GPU; Colab/Kaggle as alternative
Storage	SSD storage for dataset, checkpoints, environment & outputs

Expected Outcome & Limitations

EXPECTED OUTCOME



Accept and process user-provided textual input



Tokenize and pass text through fine-tuned MuRIL



Generate a classification prediction



Provide a confidence/probability score



Display results via a Streamlit web interface

LIMITATIONS

Dataset size, diversity & class imbalance

Difficulty with sarcasm & implicit hate

Ambiguous, context-dependent statements

Spelling variation & informal social-media language

Under-represented code-mixed variations

Possible false positives / false negatives

An automated assistance mechanism — not a replacement for human moderation.

Where the Project Can Go Next



Larger, Diverse Dataset

More multilingual and code-mixed training examples



Model Comparison

Benchmark MuRIL vs. mBERT and XLM-R



Explainable AI

Surface which linguistic patterns drove a prediction



Production API

Extend Streamlit with a dedicated model-serving API



Continuous Monitoring

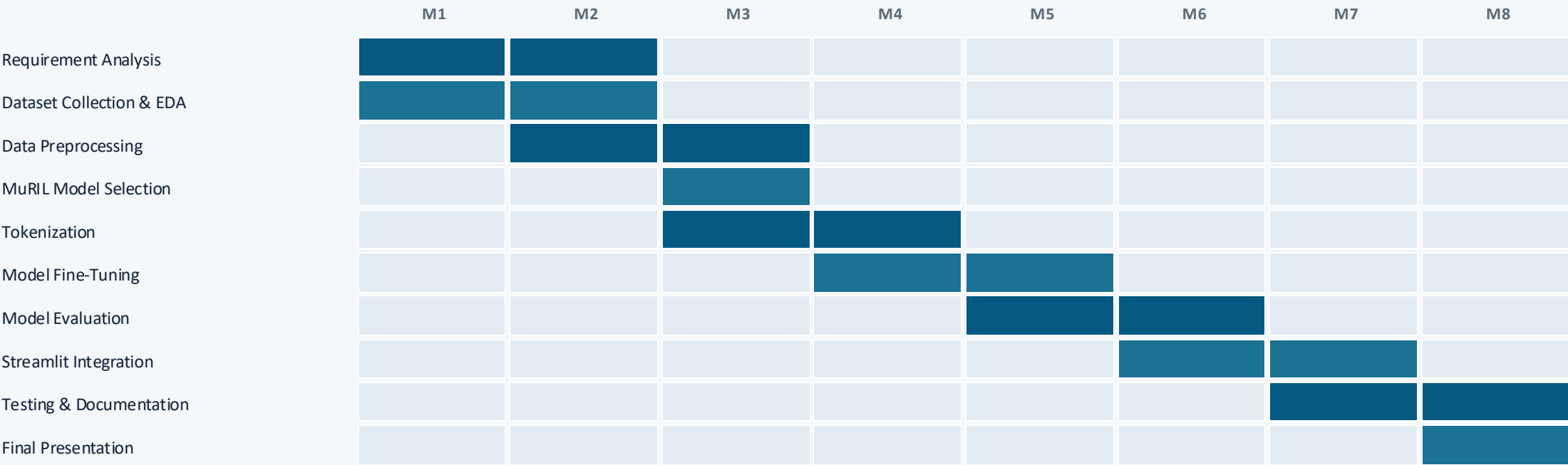
Track prediction quality as language patterns shift



Human-in-the-Loop Moderation

Route uncertain predictions to human moderators

Gantt Chart & Conclusion



Conclusion: This project adapts the pre-trained google/muril-base-cased model to hate-speech classification through transfer learning, covering dataset preparation, fine-tuning, evaluation, and a Streamlit-based real-time inference application.

References: Vaswani et al. (Attention Is All You Need) · Devlin et al. (BERT) · Khanuja et al. (MuRIL) · Wolf et al. (Transformers) · Hugging Face, PyTorch & Scikit-learn documentation